

$$(x, y) \mapsto (x^2 + c - ay, x)$$

$$G^+(x, y) := \lim_{n \rightarrow \infty} \frac{1}{2^n} \log^+ \|f^n(x, y)\|$$

Green func の 評価

$$V_R^+ = \{(x, y) \in \mathbb{C}^2 \mid |x| \geq R, |x| \geq |y|\}$$

$$B_R = D_R \times D_R; \quad D_R = \{|z| \leq R\}$$

$$R_0 = \frac{1}{2} (1 + |a| + \sqrt{(1 + |a|)^2 + 4|c|})$$

Unique pos. sol. of $R^2 - (1 + |a|)R - |c| = 0$

$$R_1 = \max \{ R_0, |a| + \sqrt{|a|^2 + 2|c|} \}$$

Unique pos. sol. of $\frac{1}{2}R^2 - |a|R - |c| = 0$

Lemma 3.1 $R \geq R_0 \Rightarrow f(V_R^+) \subset V_R^+$

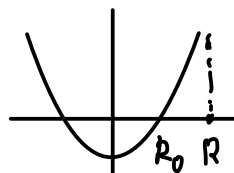
$$(x_0, y_0) \mapsto (x_1, y_1) \text{ s.t.}$$

$$|x_1| = |x_0^2 + c - ay| \geq |x_0|^2 - |c| - |a||x_0| \geq |x_0|$$

∴ By def of R_0 .

$$\Rightarrow |x_1| \geq |x_0| \geq R$$

$$\parallel \\ |y_1|$$



$$\Rightarrow (x_1, y_1) \in V_R^+$$

Prop 3.2

$$(i) (x, y) \in B_{R_0} \Rightarrow G^+(x, y) \leq \log R_0 + \log 2$$

$$(ii) (x, y) \in V_{R_1}^+ \Rightarrow G^+(x, y) \geq \log |x| - \log 2$$

Proof (i)

$$|x_1| = |x_0^2 + c - ay_0| \leq |x_0|^2 + |c| + |a||y_0|$$

$$|x_0| \geq |y_0| \xrightarrow{\text{blue}} \leq |x_0|^2 + |c| + (1+|a|)|x_0|$$

$$\text{By def of } R_0, \xrightarrow{\text{blue}} \leq 2|x_0|^2$$

$$R_0^2 > (1+|a|)R_0 + |c|$$

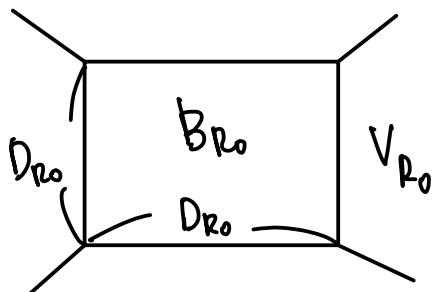
Applying Lemma 3.1 Repeatedly,

$$|x_n| \leq 2^{1+2+\dots+2^{n-1}} |x_0|^{2^n} = 2^{2^n-1} |x_0|^{2^n}$$

$$\text{Since } y_n = x_{n-1}, \quad \text{blue } |x_n| > |x_{n-1}|$$

$$\|f^n(x_0, y_0)\| \leq |x_n| + |x_{n-1}| \leq 2^{2^n} |x_0|^{2^n}$$

$$\Rightarrow \|f^n(x_0, y_0)\| \leq 2^{2^n} R_0^{2^n} \text{ on } \partial D_R \times D_R$$



$$\Rightarrow G^+(x, y) \leq \log R_0 + \log 2$$

⊖ pluri-subharmonic.

(ii) By def of R_1 ,

$$|x_1| = |x_0^2 + c - ay_0| \geq |x_0|^2 - |c| - |a||x_0| \geq \frac{1}{2} |x_0|^2$$

$$\frac{1}{2} R^2 - |a|R - |c| \geq 0$$

$$\Rightarrow |x_n| \geq \frac{1}{2^{1+2+\dots+2^{n-1}}} |x_0|^{2^n} = \frac{1}{2^{2^n-1}} |x_0|^{2^n}$$

$$\therefore \|f^n(x_0, y_0)\| \geq |x_n| \geq \frac{1}{2^{2^n}} |x_0|^{2^n}$$